Optimizing Prompts for Extracting

Bitcoin Price Predictions

from News Articles

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MOTIVATION

Noisy Signals

- Bitcoin is one of the most volatile assets in the world
- Trader base is younger, more informal language + platforms
- Emerging support technologies adapt quickly

24/7, Ever-Changing Format

- No markets–never closes
- Decentralized and scheduled events by design
- Updates can happen in hidden places

Our Goal

- Use news articles as a proxy for information stream
- Utilize prompt optimization to extract features
- Use an author as a case study to examine feasibility

Why Prompt Optimization

- Data is costly to harvest and label
- As a result, fine tuning became unrealistic
- Allows model comparison without changing weights

RESEARCH QUESTIONS

Q1

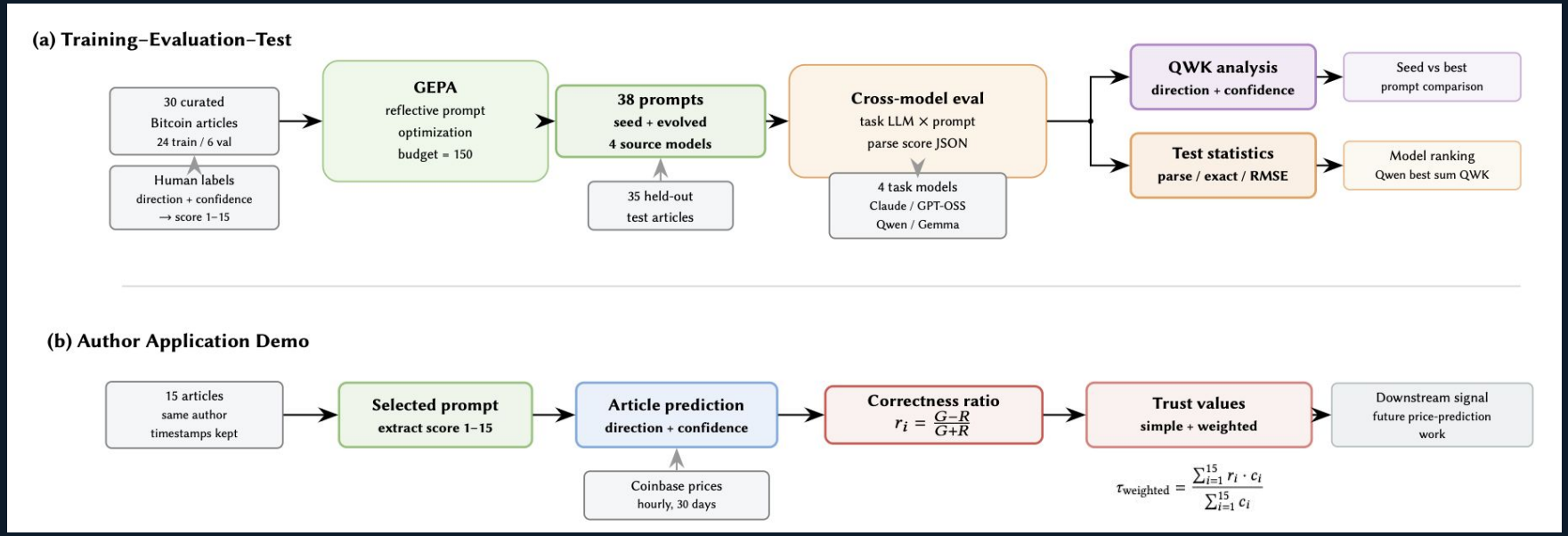
Does GEPA produce prompts that improve structured extraction of forward-looking Bitcoin sentiment compared to the shared seed prompt?

Q2

Which model best recovers human-labeled direction and confidence on the held-out Bitcoin news test set using its optimized prompt?

Pipeline: GEPA optimization ⟶ Cross-model test evaluation ⟶ QWK agreement analysis ⟶ Author-level trust demo

PIPELINE OVERVIEW



DATA & SCORING SYSTEM

24	6	35	15
Train	**Validation**	**Test**	**Author**
GEPA optimization	Prompt selection	Final evaluation	Trust demo

Unified 1–15 Score: $s = \Delta + c$ (Developed for GEPA)

Direction	Offset (Δ)	Confidence (c)	Score Range
Bearish −1	0	1 – 5	1 – 5
Neutral 0	5	1 – 5	6 – 10
Bullish +1	10	1 – 5	11 – 15

GEPA: Genetic-Pareto - a prompt optimizer that thoroughly incorporates natural language reflection to learn high-level rules from trial and error.

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg\max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x,m)\sim\mathcal{T}} \left[\mu\big(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m\big) \right].$$

Table 2: Notation for the GEPA optimization framework

Symbol	Meaning
Φ	Language-model system
Π	Collection of prompts
Θ	Model weights
x	Task input
m	Evaluator metadata
$\mathcal{T}$	Task distribution
μ	Evaluation metric
arg max	Selects the best-scoring parameters
$\mathbb{E}$	Expected or average performance

```
{
  "input": "Title: The Biggest ...",
  "answer": "14",
  "additional_context": {
    "gold_reasoning": "Compares the ..."
  }
}
```

Table 3: Partial-credit evaluator used during GEPA

Prediction outcome	Score
Exact match	1.00
Off by 1	0.75
Off by 2	0.50
Off by 3	0.25
Off by more than 3	0.00
Parse failure	0.00

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Limitation - Metric

- Partial credit based on absolute distances on the 1-15 scale
- Issue: 9 -> 11 only moves by 2 points, but crosses a clear theoretical boundary of neutral to bullish
- Future: Use an evaluation metric that gives little to 0 credit for boundary violations

GEPA OPTIMIZATION METHOD - KEY TERMS

Rollout: Running one candidate prompt on one or more training articles to see what score the task model predicts.

Budget: The maximum number of evaluator calls GEPA is allowed to spend during optimization. (In our case, set to 150)

Iteration: One GEPA loop where it selects existing prompt candidate(s), proposes a new prompt, tests it, and either keeps or discards it.

Candidate: One specific prompt version being tested by GEPA.

Pareto Frontier: The filtered set of prompts that are useful because each one is best or competitive on at least some examples, instead of only keeping the single best average prompt.

Mutate: Creating a new prompt by reflecting on one existing prompt's mistakes and rewriting it.

Merge: Creating a new prompt by combining useful parts from two existing prompts.

Minibatch: A small sampled group of training articles used to quickly test whether a new prompt seems promising.

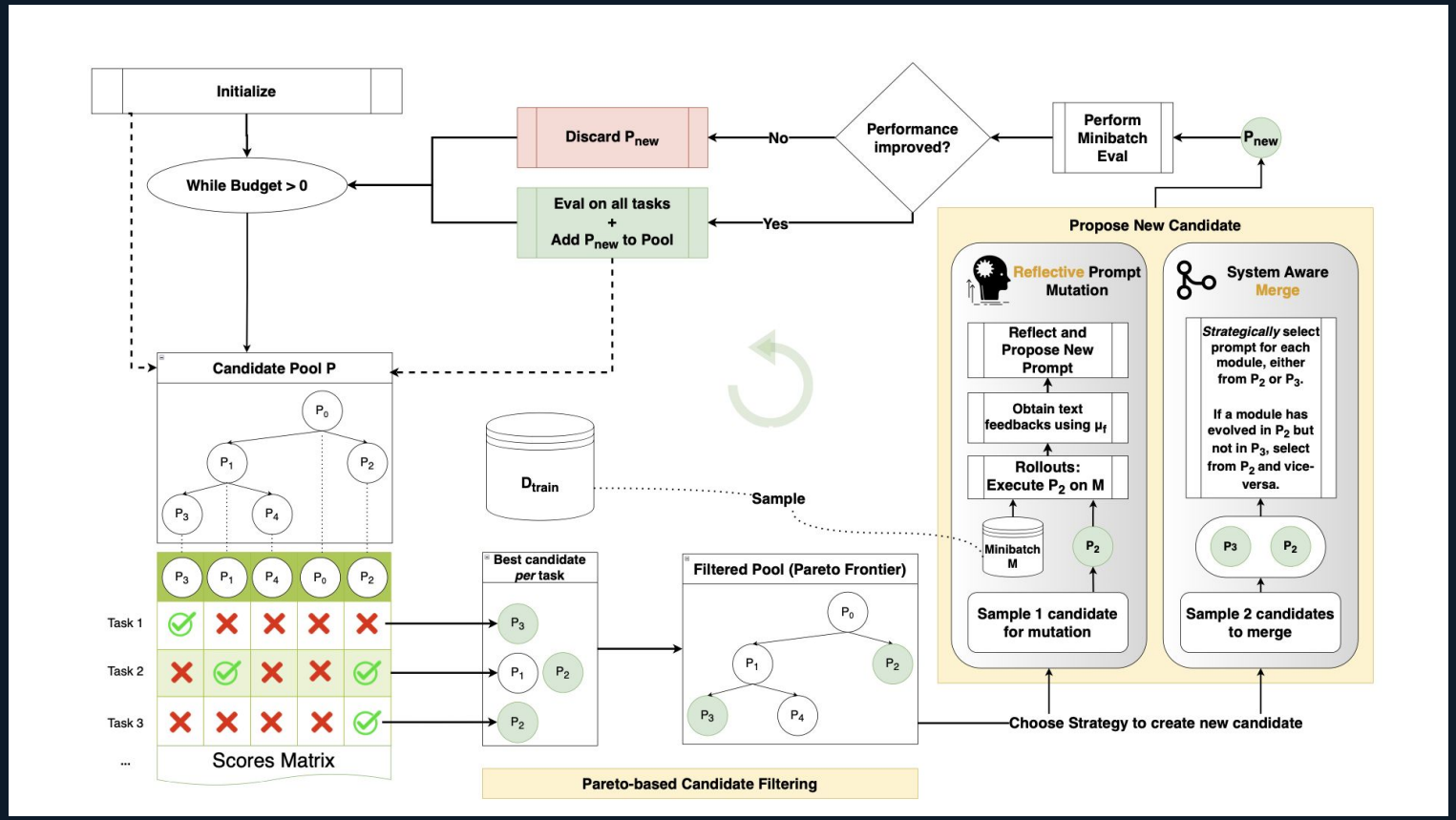
Evaluator: The scoring function that compares the model's predicted score with the human gold score and returns partial credit plus feedback (In our case, the metric shown before) .

Feedback: The text generated by the evaluator that explains whether the prediction was correct, too bullish, too bearish, or failed to parse, and including human reasoning.

Task Model: The LLM that actually reads the Bitcoin article and outputs {"score": ...}.

Reflection Model: The LLM that reads feedback and writes a revised prompt. (In our case, task and reflection model are the same)

GEPA OPTIMIZATION METHOD

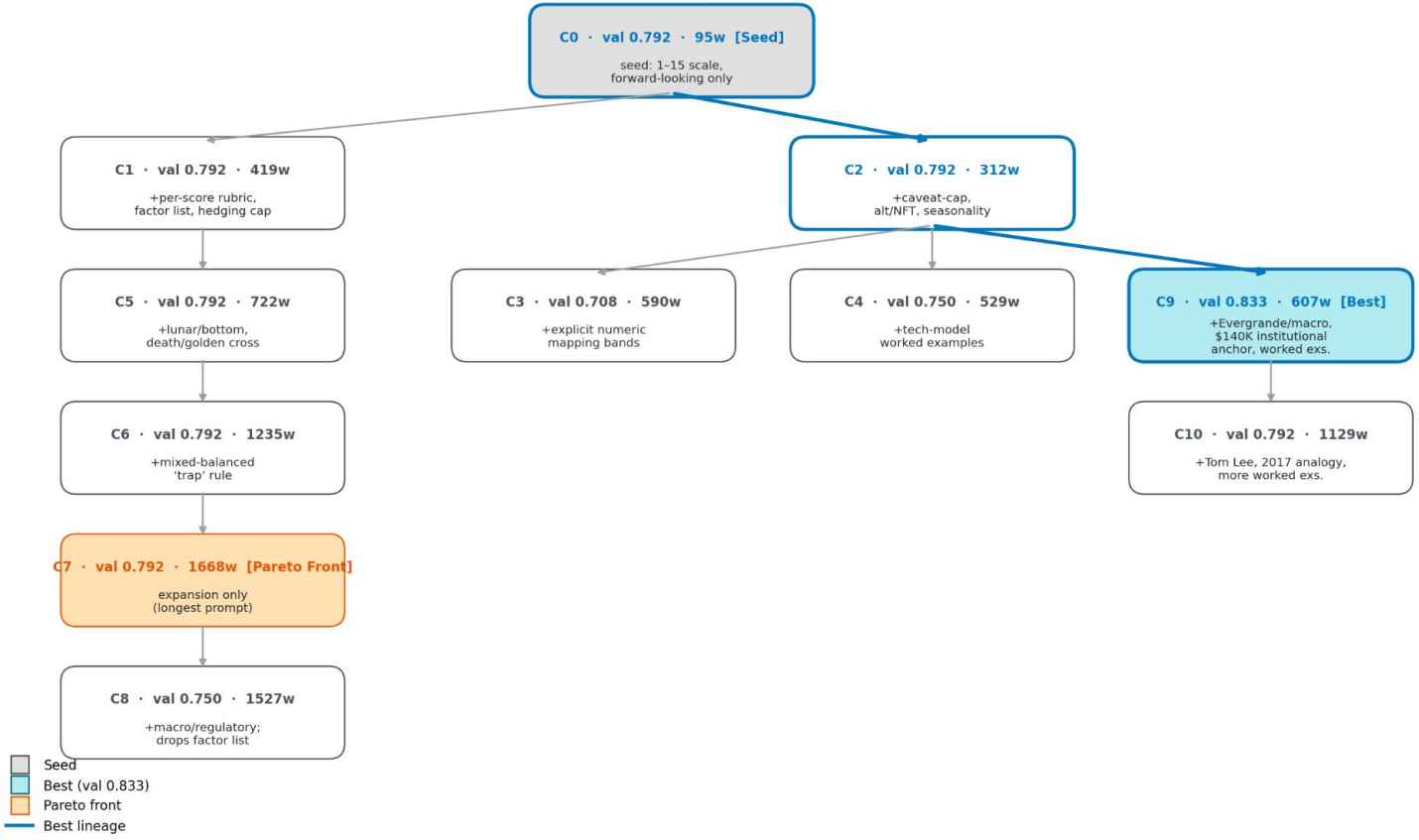


STAGE 1 — GEPA RESULTS

Source Model	Candidates	Accepted	#Words for best prompt	Growth
Claude Sonnet 4.6	11	10	615	6.68×
GPT-OSS 120B	7	6	92 (seed)	1.00×
Qwen 3.6	11	10	714	7.76×
Gemma 4 E2B	9	8	259	2.82×
Total	**38**	**34**	—	—

Source model GEPA run	Seed/base val score	Final best val score
Claude Sonnet 4.6	0.792	0.833
GPT-OSS 120B	0.875	0.875
Qwen 3.6	0.792	0.917
Gemma 4 E2B	0.667	0.833

STAGE 1 — GEPA PROMPT EVOLUTION TREE for CLAUDE SONNET 4.6



STAGE 2 — TEST EVALUATION (35 held-out articles)

Task Model	Parse Rate	Exact Acc. (on all prompt candidates)	Direction Acc. (on all prompt candidates)	RMSE
Claude Sonnet 4.6 (1330 preds)	81.3%	13%	65.5%	**2.17**
GPT-OSS 120B (1330 preds)	99.6%	22.3%	**77%**	2.85
Qwen 3.6 (805 preds)	82.2%	**24.6%**	76.0%	2.82
Gemma 4 E2B (1330 preds)	100%	23.6%	67.2%	3.57

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Dataset Size

- Only 24 training and 6 validation articles were used for GEPA
- Validation accuracy looked high: 83.3% for 3 models
- Test exact accuracy was much lower: 13–24.7%
- This suggests overfitting; the seed prompt often beat the GEPA-best prompt on the held-out test set

Stage 3 - Quadratic Weighted Kappa - a more comprehensive analysis

QWK measures agreement between two raters on an **ordered** scale while accounting for agreement that could happen by chance. It penalizes larger disagreements more heavily than small ones, so predicting confidence 3 instead of 4 is treated as much less severe than predicting 1 instead of 4.

RMSE: average size of numeric prediction errors in the original score units, with bigger errors penalized more because errors are squared.
QWK: measures agreement on an ordered rating scale relative to chance agreement.

RMSE answers “how far off are the predictions?”
QWK answers “how well do predictions agree with gold labels, accounting for label distribution and chance?”

We are doing QWK respectively to “direction” and “confidence”, then sum them together.

Stage

1. Predic
combi
across

$$\mathbf{y} =$$

$$\hat{\mathbf{y}} =$$

2. Build

$$O_{ij} =$$

5. QWK Formula

Now compute weighted observed disagreement:

$$D_O = \sum_{i=0}^{K-1} \sum_{j=0}^{K-1} W_{ij} O_{ij}$$

And weighted expected disagreement:

$$D_E = \sum_{i=0}^{K-1} \sum_{j=0}^{K-1} W_{ij} E_{ij}$$

Then:

$$\kappa_w = 1 - \frac{D_O}{D_E}$$

would look
and

$$O_{ij}$$

$$O_{ij}$$

QWK — SEED VS GEPA-BEST PROMPT

Task	Source	Seed QWK			Best QWK			Δ Best – Seed		
		Dir.	Conf.	Sum	Dir.	Conf.	Sum	Dir.	Conf.	Sum
Claude	Claude	0.908	0.254	1.162	0.912	0.049	0.961	0.003	-0.204	-0.201
Claude	Gemma	**0.926**	0.180	1.106	0.872	0.203	1.075	-0.054	0.023	-0.031
Claude	GPT-OSS	0.907	0.199	1.106	0.907	0.199	1.106	0.000	0.000	0.000
Claude	Qwen	**0.926**	0.135	1.061	0.892	0.211	1.103	-0.033	0.075	0.042
Gemma	Claude	0.702	0.480	1.182	0.710	0.388	1.099	0.008	-0.091	-0.083
Gemma	Gemma	0.789	0.329	1.117	0.792	0.453	1.245	0.004	0.124	0.128
Gemma	GPT-OSS	0.702	0.480	1.182	0.702	0.480	1.182	0.000	0.000	0.000
Gemma	Qwen	0.716	0.554	1.271	0.621	0.401	1.022	-0.095	-0.153	-0.248
GPT-OSS	Claude	0.845	0.561	1.407	0.875	0.345	1.220	0.030	-0.216	-0.187
GPT-OSS	Gemma	0.774	0.580	1.353	0.816	0.313	1.129	0.043	-0.267	-0.224
GPT-OSS	GPT-OSS	0.824	0.603	1.427	0.824	0.603	1.427	0.000	0.000	0.000
GPT-OSS	Qwen	0.824	0.579	1.403	0.912	0.238	1.150	0.088	-0.341	-0.252
Qwen	Claude	0.801	0.515	1.316	0.854	0.321	1.175	0.053	-0.194	-0.141
Qwen	Gemma	–	–	–	–	–	–	–	–	–
Qwen	GPT-OSS	0.814	**0.620**	**1.433**	0.814	**0.620**	**1.433**	0.000	0.000	0.000
Qwen	Qwen	0.814	0.587	1.401	–	–	–	–	–	–

QWK — SEED VS GEPA-BEST PROMPT

2 / 14	4 / 14	8 / 14
Improved	Unchanged	Declined

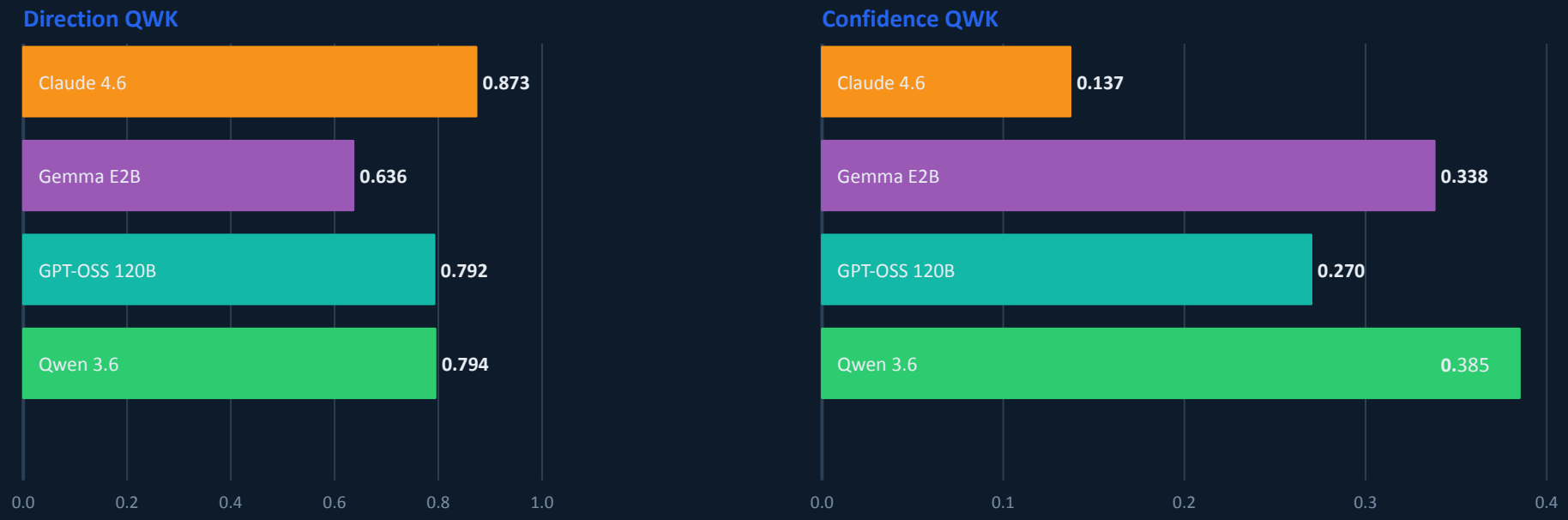
Combined QWK (Direction + Confidence) — 14 comparisons with both seed & best available

Key Pattern: Direction–Confidence Tradeoff

GEPA prompts often improved direction QWK while hurting confidence QWK.
Example: GPT-OSS + Qwen prompt → Dir +0.088 but Conf −0.341 (sum −0.252)

1.433 best sum	+0.128 Δ	−0.252 Δ worst
Qwen task + GPT-OSS seed	Gemma task + Gemma source	GPT-OSS + Qwen prompt

QWK — OVERALL MODEL PERFORMANCE



Qwen 3.6 wins combined QWK (1.178) — best confidence (0.385) + competitive direction (0.794)

Claude leads direction (0.873) but low confidence (0.137) — alignment compresses scores toward band center

Gemma: perfect parse coverage but lowest combined QWK — formatting ≠ label agreement

QWK Per Model

DISCUSSION

Why Qwen Won

Qwen's reasoning-oriented design (thinking budget, explicit chain-of-thought) helps track multiple evidence pieces to produce graded confidence scores.

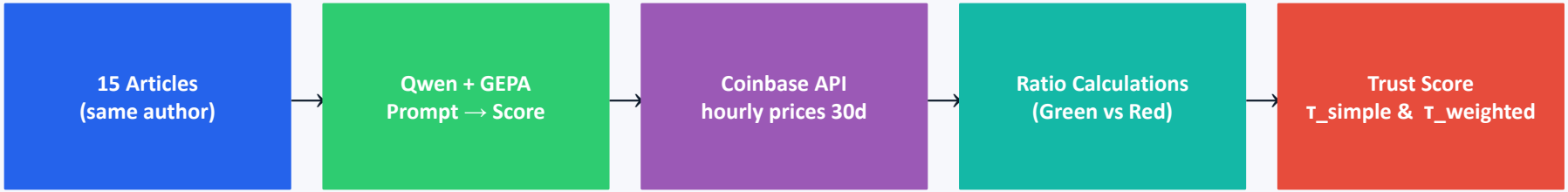
Why Claude Compressed Confidence

Heavy post-training for safety & alignment makes Claude conservative around hedged financial language, compressing scores toward the middle of each band.

Did it work?

Can the system measurably predict Bitcoin prices?

AUTHOR TRUSTWORTHINESS — METHOD



$$s_i = \Delta + c_i \tag{6}$$

where Δ is a direction-specific offset:

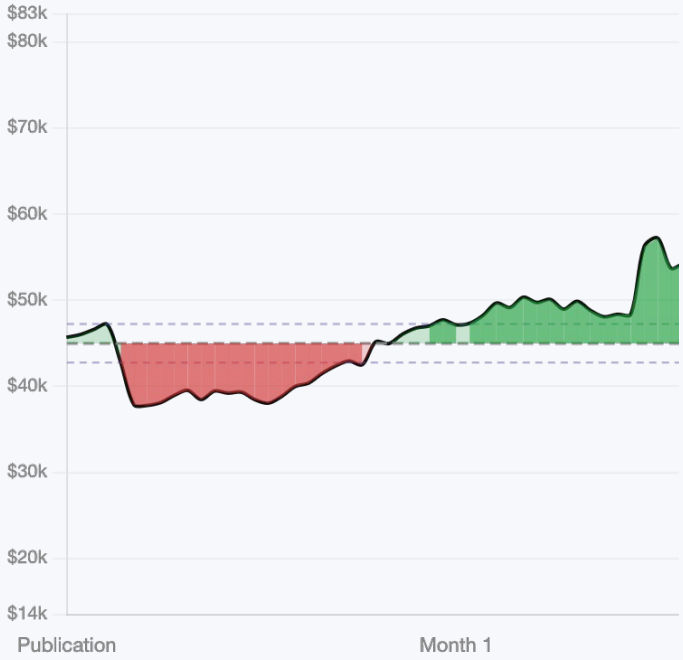
$$\Delta = \begin{cases} 0 & \text{if bearish} \\ 5 & \text{if neutral} \\ 10 & \text{if bullish} \end{cases} \tag{7}$$

and $c_i \in [1, 5]$ is the author's expressed confidence in their prediction. This gives three non-overlapping ranges:

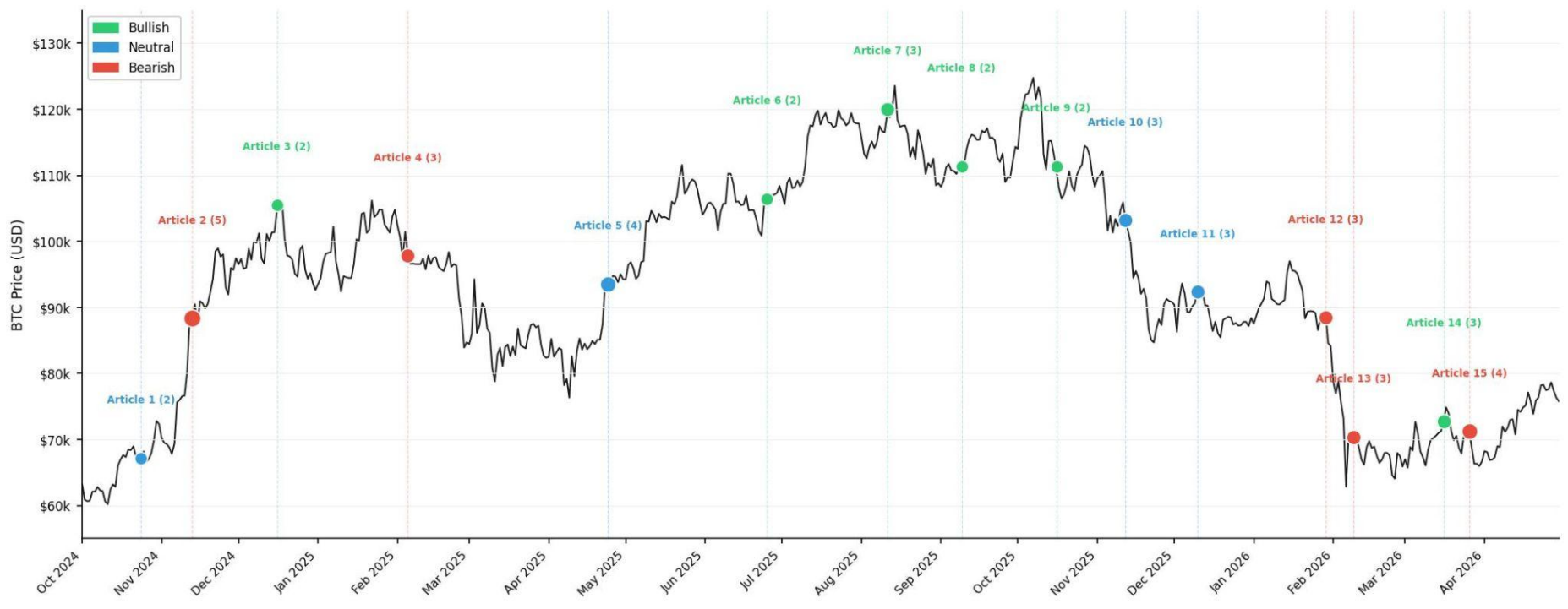
- Bearish: $s_i \in [1, 5]$
- Neutral: $s_i \in [6, 10]$
- Bullish: $s_i \in [11, 15]$

AUTHOR TRUSTWORTHINESS — METHOD

$$r_i = \frac{G - R}{G + R}$$



AUTHOR RESULTS — the Timeline



AUTHOR RESULTS — Figs 3–6 from Paper

Alice

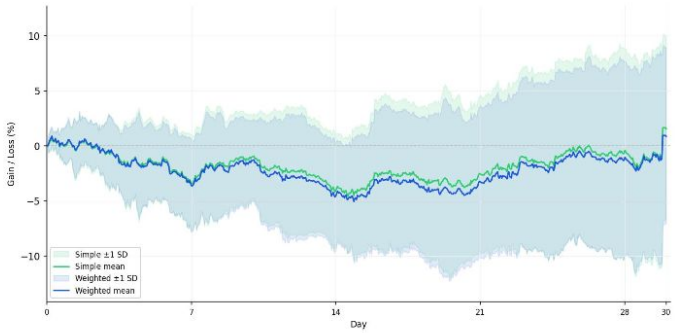


Fig. 3. 30-Day Long Position PnL on Bullish Predictions

Bob

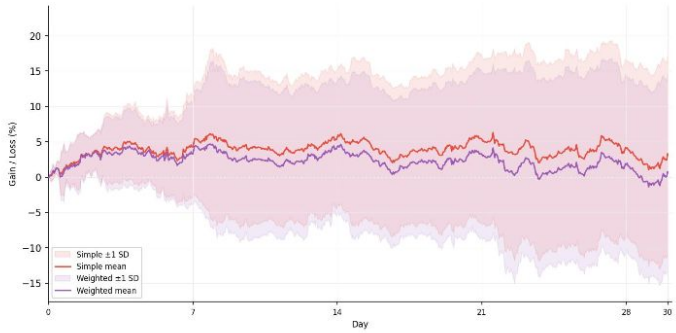


Fig. 4. 30-Day Short Position PnL on Bearish Predictions

Carlie

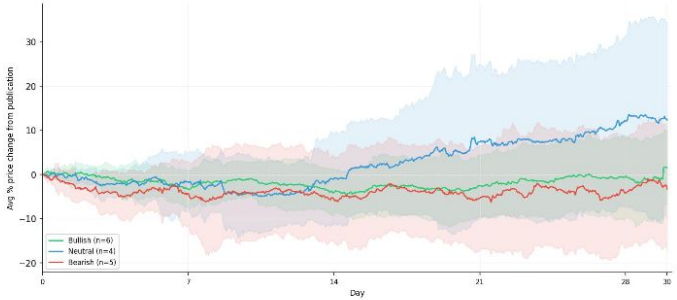


Fig. 5. Bitcoin Price Movement by Predicted Direction

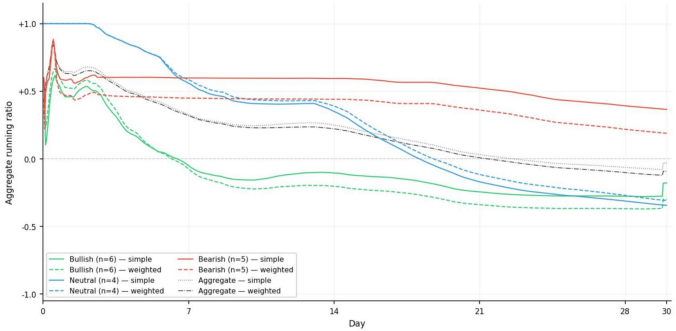
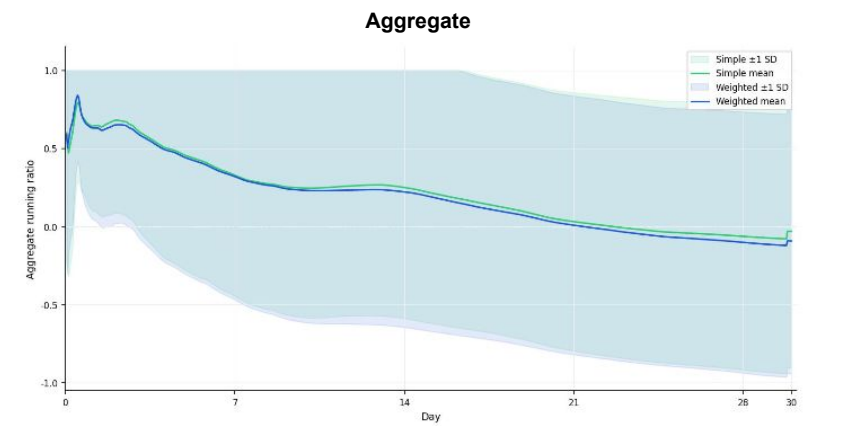
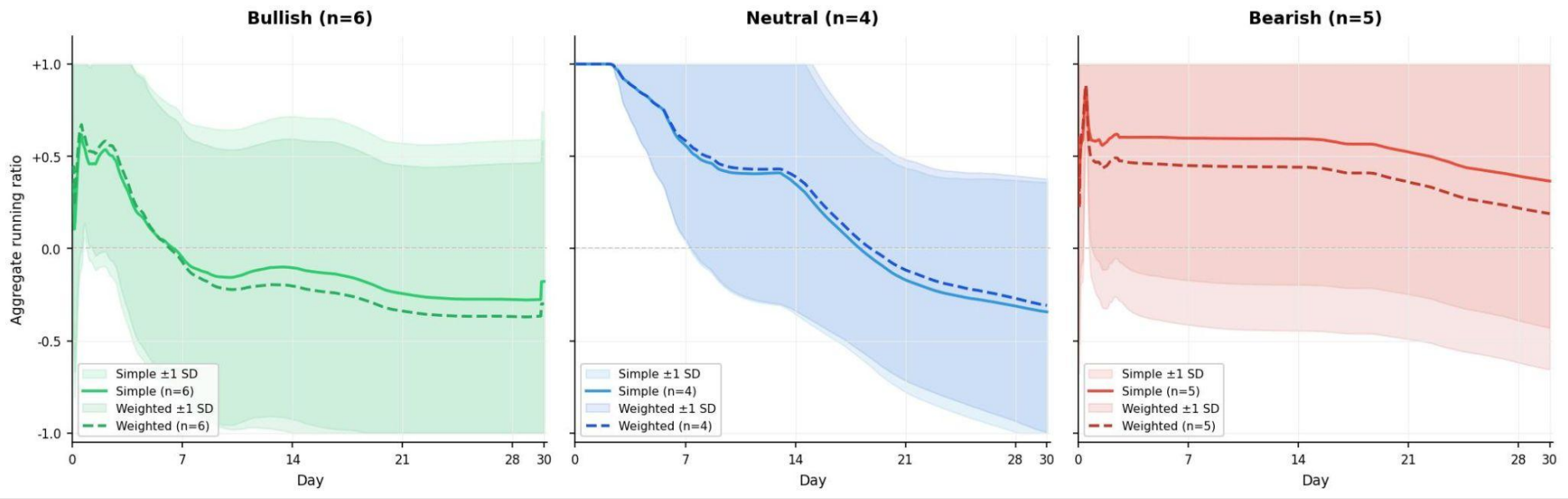


Fig. 6. Simple and Weighted Trust Ratio

Dimitri.

Dimitri.


Aggregate
Simple ±1 SD
Simple mean
Weighted ±1 SD
Weighted mean
Aggregate running ratio
1.0
0.5
0.0
-0.5
-1.0
0
7
14
21
28
30
Day


Bullish (n=6)
Neutral (n=4)
Bearish (n=5)
Aggregate running ratio
+1.0
+0.5
0.0
-0.5
-1.0
Simple ±1 SD
Simple (n=6)
Weighted ±1 SD
Weighted (n=6)
Simple ±1 SD
Simple (n=4)
Weighted ±1 SD
Weighted (n=4)
Simple ±1 SD
Simple (n=5)
Weighted ±1 SD
Weighted (n=5)
0
7
14
21
28
30
Day

3.6.5 Weighted and Unweighted Scores. Two trust scores are computed with the results.

The simple trust score is the unweighted mean ratio:

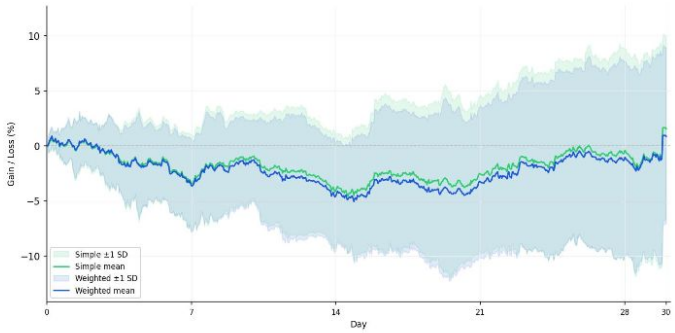
$$\tau_{\text{simple}} = \frac{1}{15} \sum_{i=1}^{15} r_i \tag{9}$$

The confidence-weighted trust score:

$$\tau_{\text{weighted}} = \frac{\sum_{i=1}^{15} r_i \cdot c_i}{\sum_{i=1}^{15} c_i} \tag{10}$$

AUTHOR RESULTS — Figs 3–6 from Paper

Alice



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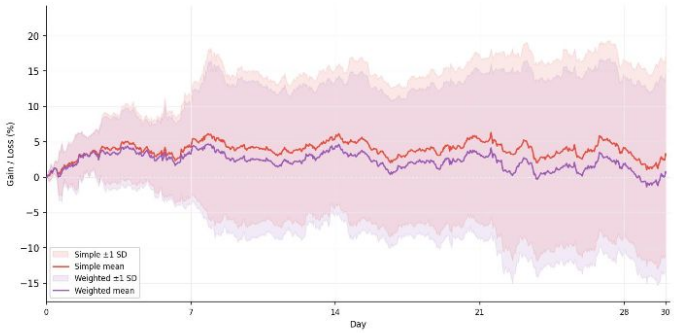
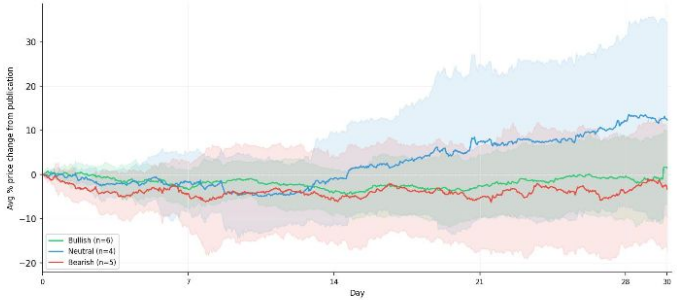


Fig. 4. 30-Day Short Position PnL on Bearish Predictions

Carlie



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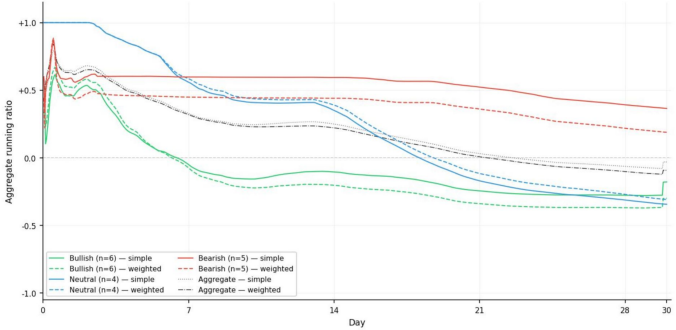

Fig. 6. Simple and Weighted Trust Ratio

Dimitri.

CONCLUSION

Q1	Mixed — GEPA generated richer domain-specific prompts, but GEPA-best did not consistently beat the seed on test QWK.
Q2	Qwen 3.6 wins overall — best confidence calibration (0.385) + competitive direction (0.794) = highest combined QWK (1.178).

Future Work

- Larger, more diverse labeled dataset
- Direction-aware training metric (penalize boundary crossings)
- Expand author trust analysis across more authors, time windows & market conditions